



Frontiers in free-energy calculations of biological systems

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In a matter of three decades, free-energy calculations have emerged as an indispensable tool to tackle deep biological questions that experiment alone has left unresolved. In spite of recent advances on the hardware front that have pushed back the limitations of brute-force molecular dynamics simulations, opening the way to time and size scales hitherto never attained, they represent a cogent alternative to access with unparalleled accuracy the thermodynamics and possibly the kinetics that underlie the complex processes of the cell machinery. From a pragmatic perspective, the present review draws a picture of how the field has been shaped and invigorated by milestone developments, application, and sometimes rediscovery of foundational principles laid down years ago to reach new frontiers in the exploration of intricate biological phenomena. Through a series of illustrative examples, distinguishing between alchemical and geometrical transformations, it discusses how far free-energy calculations have come, what are the current hurdles they have to overcome, and the challenges they are facing for tomorrow. © 2013 John Wiley & Sons, Ltd.

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INTRODUCTION

About 35 years ago, the first molecular-dynamics simulation of a small protein, the bovine pancreatic trypsin inhibitor, by McCammon et al.,¹ marked a turning point in computational structural biology. Although molecular dynamics had been used increasingly since the seminal simulation of Alder and Wainwright,² 20 years earlier, its application to biological macromolecules was not only novel, but it also quickly opened new vistas for the exploration of the cell machinery by means of numerical experiments. As it just as rapidly became apparent that capturing the complexity of biological phenomena was subservient to the available computer power, it became equally obvious that given the timescales amenable to molecular dynamics, observation of rare events would require more than brute-force simulations. It was true

then and, to a large extent, it still remains so today, notwithstanding the extraordinary dilation of time and size scales that high-performance parallel computing made possible in recent years.

The complexity of the modeled objects combined with the appreciable time span of the processes of interest (see Figure 1)—however short on the biological timescale, almost naturally imposes that alternate avenues be examined to sample the underlying, characteristically slow degrees of freedom. Less than a decade after the first molecular-dynamics simulation of a small protein, free-energy calculations of biological systems have emerged as an essential tool to achieve this goal.³ Metaphorically, they can be viewed as *computational tweezers* utilized to anatomize and explore the relevant order parameters, either physical or nonphysical, along which sampling is performed in a preferential fashion.

The primary objective of the present contribution is not to provide the reader with an exhaustive review of the maelstrom of free-energy calculations published heretofore, but to rather paint a chronological fresco of the main developments that have shaped the field, accompanied by a few cogent, sometimes self-serving illustrations. In lieu of dwelling on their

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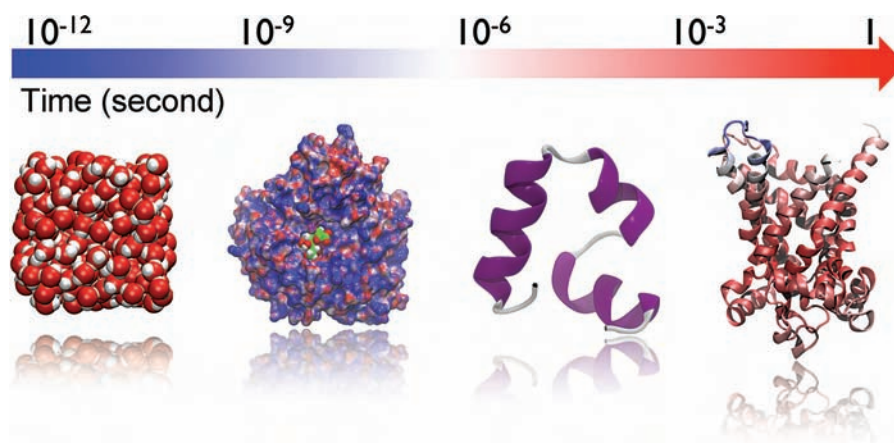


FIGURE 1 | Typical timescales spanned by biological processes of the cell machinery. From left to right—Translational and orientational relaxation of water on the picosecond timescale; Spontaneous binding of zanamivir or Relenza,⁴ an inhibitor of neuraminidase in the A/H1N1 virus; Folding of the villin headpiece subdomain, a fast-folding protein formed by 35 amino acids; and Conformational transition in the mitochondrial transporter of adenosine di- and triphosphate.

theoretical underpinnings, which can be found elsewhere at different levels of detail,^{3,5} free-energy calculations of biological systems are introduced from a pragmatic perspective—how to choose the most relevant method to tackle the problem at hand? To address this question, a distinction is being made between two classes of free-energy transformations, namely those of alchemical nature and those of geometrical nature, exemplified by a handful of applications that epitomize the frontiers reached in this research area. The review closes with an outline of the ongoing challenges in the field, alongside with the novel directions that are currently being explored.

WHAT IS THE BEST METHOD FOR A GIVEN PROBLEM?

At the applicative level, free-energy calculations of biological systems can be generally distinguished in terms of geometrical and alchemical transformations. Although the latter exploit the malleability of the potential energy function and the virtually infinite possibilities of computer simulations to transform between chemically distinct states, the former embrace positional, orientational, and conformational changes in macromolecules and complexes thereof. At the methodological level, such transformations are achieved using a variety of approaches, which, roughly speaking, can be subdivided into four different classes— (1) probability distributions and histograms,^{6,7} (2) perturbation theory,^{8,9} (3) nonequilibrium work,¹⁰ and (4) gradient-based methods¹¹. Given the resemblance of some of the nu-

merical schemes pertaining to these subdivisions, their common denominators, and the possibility to interconnect them, the latter classification is somewhat arguable and, hence, not unique.

A recurrent question in the realm of free-energy calculations is the choice of the most relevant approach to tackle the problem at hand—a conundrum, which could be reworded in terms of the best-suited, cost-effective method to obtain a reliable answer. To a large extent, this question can be addressed by considering the actual nature of the transformation, alchemical, or geometrical, and subsequently by deciding which one, among the aforementioned four classes of methods, is more likely to yield the desired free-energy change at a minimal computational effort.

The free-energy cost for protein insertion in the cell membrane provides a rather cogent illustration of how the choice of the method impacts our perception of the biological problem and how the latter ought to be handled numerically. Contrary to chemical intuition, experiment reveals that the energetic penalty incurred in the partitioning of a charged arginine amino acid amounts to only 2 to 3 kcal/mol,¹² when computer simulations predict a free-energy cost about fivefold larger.^{13,14} In an attempt to reconcile the discrepant estimates from theory and experiment, the thermodynamics of insertion has been measured by translocating from water to the hydrophobic core of a lipid bilayer, an arginine residue borne by an integral, pseudo-infinite poly-leucine-helix. The potential-of-mean-force calculation, wherein the background α -helix is translated along the normal to the interface, yields a net free-energy difference of about 17 kcal/mol,¹⁵ in line with previous theoretical estimates,

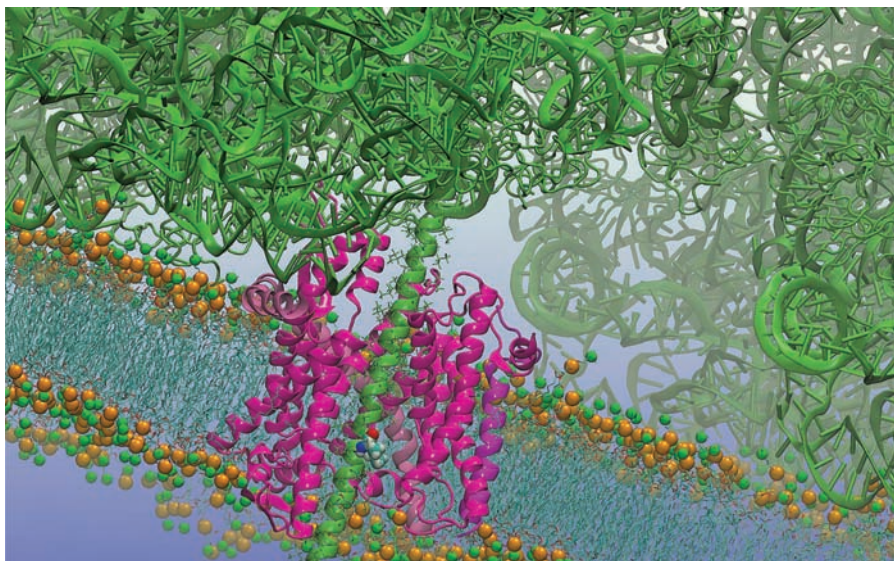


FIGURE 2 | Intricacy of the biological objects forming the cell machinery. Shown in this artistic composite view is the translocon (magenta), an integral membrane protein, to which the ribosome (green) is anchored. The exit tunnel of the latter forms with the former a continuous translocation channel along which the synthesized peptide chain is conveyed. Free-energy calculations help understand how the hospitable environment of the translocon not only reduces the cost for translocating charged amino acids—e.g., an arginine residue depicted in van der Waals spheres, but also the gain for translocating hydrophobic ones, resulting in a compressed hydrophobic scale.¹⁶

as the environment of the arginine side chain changes from the bulk aqueous medium to the midst of the model membrane.

This free-energy calculation, however, raises a methodological issue. Because it focuses primarily on the end points of the free-energy profile, an expensive, cumbersome determination of a potential of mean force, inherently poorly converging on account of the slow relaxation of the aliphatic lipid tails in response to the moving integral helix, is not necessarily justified—inasmuch as knowledge of the free-energy change across the complete translocation pathway is evidently of lesser interest. Under these premises, a perturbative approach, whereby an arginine residue found within the pseudo-infinite helix and lying in water, far from the surface of the bilayer, is mutated into leucine, whereas a leucine residue lying amidst the hydrophobic core of the membrane is transformed into arginine, appears to be somewhat more effective (see Figure 2).¹⁶

ALCHEMICAL FREE-ENERGY CALCULATIONS

Reminiscent of the unattainable aspiration of the proverbial alchemist to transmute base metal into gold, alchemical free-energy calculations^{3,17} are aimed at transforming chemical species by virtue of

the malleability of the potential energy function and the possibility to alter either the intermolecular interaction as a whole, or the individual force-field parameters. At the algorithmic level, transformation between states can be carried out using two distinct routes. In the single-topology paradigm,¹⁸ a common denominator of the two end states is sought and the free-energy change is computed by modifying selectively the nonbonded parameters of the force field—the charges and the van der Waals radii and well depths are expressed as a linear combination of their end-point values by means of a general-extent parameter. Alternatively, in the dual-topology paradigm,¹⁹ the reference and the target states of the alchemical transformation are defined concomitantly, yet without interacting. The intermolecular interaction of the perturbed moiety with its environment then becomes a linear combination of the end-state potential energy functions through the same general-extent parameter. Although the two paradigms may be used interchangeably, under certain circumstances, one might be better suited than the other—e.g., the perturbation induced by the protonation or deprotonation of a titratable amino acid, for instance in constant-pH simulations,^{20,21} would admittedly be better handled by the single-topology paradigm, wherein a single hydrogen atom either appears or disappears, than by the dual-topology, which supposes the duplication of the side chain.

Conceptually, alchemical transformations are by tradition associated to perturbation theory,^{8,9} probably because their target state often constitutes a perturbation, small or large, of the reference one. Free-energy perturbation calculations are, therefore, commonly employed to model chemical alterations within molecular assemblies, in general, and site-directed mutagenesis experiments, in particular. However, well suited to tackle alchemical transformations, free-energy perturbation is not the sole option available to the modeler—gradient-based approaches, like thermodynamic integration,¹¹ represent a robust alternative toward reliable free-energy difference,²² albeit there is by and large no fundamental reason why one route ought to be preferred over the other. From the point of view of the practitioner, it is actually more a matter of long-standing habit and possibly dogma, as the two classes of methods have coexisted over decades without any compelling demonstration that one is sufficiently superior to justify abandoning the other. As will be discussed hereafter, free-energy perturbation, however, offers a convenient framework to measure with utmost rigor both the precision and the accuracy of the free-energy estimate.

A Tool to Address Host–Guest Chemistry Questions

Among the many applications that fall into the category of alchemical transformations, host–guest chemistry arguably occupies the most prominent position.²³ Host–guest chemistry ought to be construed here in the common acceptance of the term, the problem at hand being the determination of the affinity of a guest molecular specie for a given host. Taken in its most general and barest sense, solvation free-energy calculations^{24–26} can be seen as the prototypical case of host–guest chemistry, wherein the host is formed by the surroundings—i.e., the solvent, of the guest. Just like the first molecular-dynamics simulation of a small protein by McCammon et al.¹ opened new perspectives in computational structural biology, the seminal hydration free energy of Jorgensen and Ravimohan¹⁸ in 1985 laid the groundwork for the alchemical free-energy calculations to come. In fact, the golden age of such calculations, in the late eighties, early nineties, has been the theater of many theoretical investigations that have improved our understanding of molecular recognition and association phenomena. Barring a number of bold attempts to tackle the intricate problem of absolute or standard binding free energies,^{27–29} most endeavors were limited to the determination of relative quantities,^{30–34} trying to reconcile experimental thermodynamic mea-

surements with new, microscopic insights gained from numerical simulations—to this date, publishing computational estimates bereft of experimental support remains, in most reputable venues, at least as challenging as performing the free-energy calculation itself.

Good Practices, Guidelines, and Recommendations

The difficulty to report theoretical predictions of binding affinities can be rationalized in large measure by the haphazard reliability of free-energy calculations until recently. Over the past years, much effort has been channeled in an enhanced characterization of the available methodology and treatment of the statistical information accrued in the simulations, modifying our somewhat subjective perception of free-energy methods as black boxes. From this impetus, a number of guidelines and recommendations have emerged,³⁵ notably in the context of perturbation theory—albeit generalization to other classes of approaches is possible and, by and large, straightforward. An important aspect of free-energy calculations is the provision of error bars associated to the theoretical estimates, which encompass a statistical component—i.e., the variance, and a systematic component—i.e., the bias from the finite-length nature of the simulation.³⁶ Measuring the reliability of computed free-energy differences supposes that the variance of the underlying probability distributions characterizing the transformations, forward and backward, is appreciably small—in other words, should the estimated free-energy change be meaningless on account of insufficient sampling and poor convergence, the error bars are likely to be equally meaningless.

The concept of bidirectional transformation that has just been mentioned provides yet another cogent illustration of the compartmentalization of research areas. As early as 1976, Bennett³⁷ demonstrated in a landmark publication that combination of the statistical information accrued in a forward and a backward transformation yields a maximum-likelihood, lowest-variance estimator of the free energy. Surprisingly enough, among the many free-energy calculations reported in their golden years, only a fraction of them make use of the so-called Bennett acceptance ratio, most estimates being provided by brute application of either free-energy perturbation or thermodynamic integration. It is equally interesting to note that in their 1967 pioneering article, McDonald and Singer³⁸ had actually fashioned the framework for exploiting the wealth of information

supplied by probability distribution functions. Only recently has it been accepted that the overlap between the underlying distributions of forward and backward transformations ought to be examined carefully to assess the reliability of the free-energy estimates. It has also been recognized that combination of the statistical data accrued in bidirectional simulations, either in the rudimentary simple-overlap sampling³⁹ algorithm, in the standard Bennett acceptance ratio,³⁷ or in its more sophisticated version, the multistate Bennett acceptance ratio,⁴⁰ was a prerequisite toward more reliable and more efficient free-energy calculations.

By contrast, staging, or stratification, has been a long-known concept,⁴¹ widely utilized—albeit often in an empirical manner, without proper justification. Yet, staging constitutes a general method to reduce the variance^{5,42} and enhance the aforementioned overlap between the probability histograms obtained in each stratum of the reaction pathway. Fine-tuning of the variance is achieved by adapting the increment of the general-extent parameter introduced in the potential energy function to transform between states.³⁵ Finding an optimal balance between the number of strata and the sample size per stratum remains, however, a topical question—finer staging does not necessarily warrant lesser sampling at each stage. In fact, regardless of the stratification scheme, convergence of the free-energy calculation is subservient to the appropriate sampling of the slowest degrees of freedom of the molecular assembly of interest. In this respect, mathematical tools like the Kullback–Leibler divergence,⁴³ which allows the sample size of a forward transformation to be related to the dissipated work of its backward counterpart, provide a rather cogent explanation why convergence of insertion free-energy calculations is more rapidly attained than for the opposite deletion process.⁴⁴

In the applicative field of alchemical transformations, a common-sense recommendation is the decoupling of Coulomb and van der Waals interactions to eliminate possible instabilities in the molecular-dynamics trajectories, which might arise from the creation or annihilation of charged particles. Furthermore, such transformations ought to be carried out systematically with a soft-core potential^{45,46} designed to avoid singularities in the van der Waals term upon creation of atoms in a locus already occupied by elements of the molecular assembly, e.g., solvent molecules. In host–guest chemistry applications, free-energy calculations aimed at reproducing binding affinities sometimes make use of harmonic potentials, in particular to tether the guest. As will be discussed hereafter, contribution of these geomet-

rical restraints should be accounted for to close the corresponding thermodynamic cycle.

The Long-Standing Protein-Ligand Problem

A special case of host–guest chemistry and an ideal candidate for alchemical transformations is the protein–ligand problem—a selling argument for free-energy calculations beyond academic doors, notably in the realm of pharmaceutical sciences.¹⁷

First and foremost, it is legitimate to wonder in what way can these numerical simulations improve our current understanding of how a small molecule, possibly a therapeutic agent, binds to a protein of interest? Protein–ligand free-energy calculations are in principle able to address two distinct questions, namely (1) predict the binding affinity of a drug for a protein and, by extension, rank drug candidates according to their relative affinity toward that protein, and (2) predict the change in the binding affinity of a given drug upon point mutation of the host protein. Beyond the mere numerical aspect of these simulations aimed at supplying the thermodynamic picture of protein–ligand association in the form of a single number—the binding affinity, either relative^{30–34,47–51} or absolute,^{27–29,52–59} free-energy calculations offer additional valuable insights by shedding light into the spatial arrangement of the complex and modification thereof in response to a chemical perturbation of either the host or the guest.

Second, equally legitimate is the question of where we stand after nearly three decades since the seminal, prototypical free-energy calculation of Tembe and McCammon⁶⁰ with a model protein–ligand assembly. Have free-energy calculations really come of age to tackle on a routine basis the problem of protein–ligand association and recognition?⁶¹ Browsing through the plethora literature, it is quite intriguing that, notwithstanding a far better understanding of the methodology available today, together with access to appreciably longer timescales, progress in the field still appears slower than one would have foretold in light of the many success stories published in the golden years of free-energy calculations.

With two decades of hindsight, it is fair to acknowledge that a number of these early successes were somewhat fortuitous and, through overadvertisement, tend to conceal many undocumented failures. As a robust theoretical framework now forms the foundations of the best practices and as simulations can be pushed over hitherto inaccessible length scales, overlooked convergence issues become more apparent, alongside with shortcomings of empirical force fields—two of the likely culprits that

still prevent free-energy calculations to be performed beyond academic walls. Assuming that convergence issues—which encompass metastabilities and quasi-nonergodicity scenarios, can be addressed, the master key to the routine use of free-energy calculations outside institutional laboratories, notably in the pharmaceutical industry, is, thus, more a matter of force-field development than a computational hurdle.

The crude nature of minimalist, multipurpose potential energy functions, originally devised for handling proteins and nucleic acids is often incompatible with the level of description expected for small drugs binding to proteins. Many physical contributions that participate in recognition and association phenomena are, indeed, absent from such force fields, which, for the most part, rely upon stringent approximations—e.g., pairwise additivity, and, hence, neglect through-space induction effects, of far-reaching consequences for charged ligands. Considerable effort has been invested in recent years to revamp and extend the popular macromolecular potential energy functions in an attempt to provide a better representation of basic to complex functional groups found in organic compounds.^{26,62,63} It would be desirable that these generalized force fields be probed in a systematic fashion^{64,65} in a series of protein-ligand free-energy calculations, using not only a common, large training set of small molecules, but also a common sampling strategy. Such an endeavor can only be conceived in the context of a community effort, thankless in essence, albeit eminently necessary to pinpoint the weaknesses of the tools at our disposal.

Furthermore, to be truly useful in an industrial setting, free-energy calculations ought to be easy to design and set up, possibly in an automated fashion, not prohibitively expensive to run and painlessly analyzable to assess their reliability.^{17,66,67} In many respects, the latter requirements are antinomic with the prescriptions of good practices³⁵ outlined previously. They also suppose an appreciable human effort that precludes the use of free-energy calculations as a high-throughput virtual-screening tool.

Still, claiming that the dangling question of addressing protein-ligand association by means of free-energy calculations has remained without tangible advances over the past years would be rather unfair. To a large extent, although foundational ideas had been put forth earlier,^{68–71} Woo and Roux⁷² changed our perception of how these simulations ought to be run by developing a theoretical framework based on the introduction of geometrical restraints, in particular for the determination of standard, or absolute,

binding affinities—though it might be desirable to enforce similar restraints, in the form of harmonic potentials, for the computation of relative binding free energies. From a historical perspective, it is interesting to note that though altering the potential energy function in the spirit of relative free-energy calculations had been a widespread practice since the mid-eighties, only a handful of attempts were made to tackle the issue of standard states—a prerequisite for the computation of absolute binding constants, in either alchemical,²⁷ or geometrical⁵² transformations. Dormant over a decade, the question resurfaced in the mid-nineties,^{68–70,73} spurring conceptual ideas that eventually led to the development of a formal framework for the determination of standard binding free energies.

The practical benefit of including geometrical restraints is rationalized by the loss of configurational entropy that accompanies protein-ligand association and that is not easily captured in numerical simulations.^{68,70} In fact, when the substrate is decoupled alchemically from the host protein,⁶⁹ it becomes free to wander away from the binding site. Over the timescales amenable to molecular dynamics, the chance that the ligand forms the correct network of hydrogen bonds and van der Waals contacts with the protein upon recoupling in the backward transformation is left to serendipity. The long-time known problem of the floating ligand violates thermodynamic microreversibility and further explains why so many studies report only forward, decoupling transformations.

Likewise, from the standpoint of geometrical transformations, it is appealing, albeit deceitful, to believe that standard binding free energies are easier to measure by dissociating the protein-ligand complex—possibly through nonequilibrium work pulling experiments,⁷⁴ than by simulating the reverse, association process. Oversimplifying protein-ligand binding in terms of a one-dimensional potential of mean force⁷⁵ is tantamount to hypothesizing positional, orientational, and conformational averaging. Either from an alchemical perspective, or from a geometrical one, nothing can be further from the truth than this assumption. By restricting the configurational space of the ligand to its native position, orientation, and conformation in the complex with the protein, by means of harmonic potentials acting on the relevant polar coordinates, Euler angles and root mean-square deviation with respect to the reference structure, and accounting for these geometrical restraints in separate free-energy calculations, both in the bound state and in the free state—for which positional and orientational terms can be

derived analytically, closure of the complete thermodynamic cycle delineating reversible association is ensured.^{55,72}

Somewhat extraneous to the field of free-energy calculations, brute-force molecular dynamics simulations have emerged as yet another tool for dissecting protein-ligand recognition and association phenomena. By harnessing the considerable computational power of large arrays of cores harbored in graphics processing units,⁷⁶ or by resorting to dedicated architectures,⁷⁷ tackling the spontaneous binding of a ligand to a protein is becoming feasible at the price of microsecond-time-scale simulations aimed at mapping the association pathway delineating the formation of protein-ligand complex. Two investigations published recently provide a rather convincing illustration of the current frontiers of molecular dynamics and of what can be inferred from brute-force simulations. The first work,⁷⁸ emanating from D.E. Shaw Research, delves into the mechanism whereby known therapeutic agents associates with the β_1 - and β_2 -adrenergic receptors, two membrane proteins pertaining to the family of G protein-coupled receptors, a class of proteins involved in a large variety of processes of the cell machinery and in the limelight in 2012 with two Nobel prizes for their study. Although isolated long trajectories undoubtedly help understand how binding proceeds from the recognition of the drug by the protein, its entry into the internal cavity, to the moment it is eventually locked in place, they also admittedly provide little information about the thermodynamics of the phenomenon of interest—especially given the timescale spanned by the latter, which ought to be put in perspective with the size of the molecular assembly.

By contrast, in their reconstruction of the complete sequence of events that marks the binding of the prototypical benzamidine to the enzyme trypsin, Buch et al.⁷⁹ generated about 500 independent one-hundred-nanosecond trajectories, about one-third of which were reactive and lead to the formation of the enzyme-inhibitor complex. On the basis of this collection of simulations, not only thermodynamic, but also kinetic data were inferred, quantifying an otherwise qualitative picture of the binding process. With an accuracy comparable to that of conventional free-energy calculations, the standard binding affinity was determined, albeit at the price of a heroic computational effort of about 50 microseconds, that is approximately 500 times what a calculation based on perturbation theory would cost, thus, calling into question the use of brute-force molecular dynamics for addressing thermodynamic questions beyond a mere proof of concept.

GEOMETRICAL FREE-ENERGY CALCULATIONS

In a general sense, geometrical free-energy calculations refer to any transformation that does not involve a chemical alteration of the molecular system of interest—in a nutshell, any transformation that is not of alchemical nature. Such calculations embrace transport phenomena, recognition, and association processes, as well as conformational equilibria. Their common denominator is the definition of a reaction coordinate;^{3,5} the free-energy change, on the other hand, can be measured using either one of the four classes of methods outlined previously.

What is a Good Reaction-Coordinate Model?

Semantically, the vocabulary reaction coordinate utilized to describe the variable along which the variation of the free energy is determined is somewhat ambivalent. Although it is commonly employed to characterize the order parameter chosen to address the problem at hand, the reaction coordinate generally refers to a unique mathematical object on $\mathbb{R}^{3N}$, which defines the minimum-free-energy pathway connecting the reference state to the target state of the transformation. The reaction coordinate is often associated to the concept of committor,⁸⁰ the value of which at a given point in collective-variable space is the probability that the trajectory will reach the target state before it returns to the reference state. The minimum-action transition pathway is foliated into successive surfaces of equal commitment probability, or isocommittors.⁸¹ In general, commitment probabilities, per se, are neither useful, nor manageable, to the extent that for obvious practical reasons, the reaction coordinate is modeled in terms of a reduced set of variables of relevant collectivity—from a simple Euclidian distance to a root mean-square deviation with respect to a reference structure, or to an eigenvector.⁸²

The search for an optimal representation of the reaction coordinate is a topic in its own right, the discussion of which falls outside the scope of the present contribution. Still, considering the complexity of the process of interest, the modeler is frequently forced to resort to a reduction of dimensionality and use the geometric functions that have been made available toward this end. Under most circumstances, intuition guides the choice of the collective variables and combination thereof—albeit intuition is not necessarily synonymous with computational efficiency.

For instance, ion transport through a synthetic membrane channel consisting of a series of stacked

cyclic peptides could *a priori* be well described by means of a model reaction coordinate formed by the normal of the lipid-bilayer surface.⁸² In practice, however, ions moving in the hollow tubular cavity tend to graze the walls of the latter, binding strongly to the polar moieties of the cyclic peptides, thereby thwarting longitudinal diffusion. In spite of the bias applied along the normal to the water-membrane interface, exploration of the pathway connecting the two mouths of the channel appears incomplete, slowed down by lateral diffusion in the plane perpendicular to the model reaction coordinate. This paradigmatic example illustrates how an apparent fast degree of freedom—i.e., translation of the ion, coupled to another, slow degree of freedom—i.e., binding to the peptide nanotube, results in an effectively hampered longitudinal diffusion.

Under these premises, two options can be considered to circumvent the latter difficulty. Over the timescale amenable to molecular dynamics, a naïve one-dimensional order parameter is evidently insufficient to guarantee that at any given value of the latter, the other degrees of freedom will be adequately sampled—especially if slow collective modes are strongly coupled to the model reaction coordinate.

Increasing dimensionality constitutes one possible route, as alluded to in the above example of ions diffusing through a synthetic membrane channel.^{82,83} Another cogent illustration is supplied by the toy model formed by decaalanine,⁸² a short, 10-residue peptide, the reversible unfolding of which has been examined in a number of computational studies.^{84–86} Notwithstanding the questionable relevance of end-to-end distances far below that characteristic of an α -helix, in particular in the context of peptide folding, exploration of very compact secondary structures sheds light on the shortcomings of exaggeratedly coarse-grained model reaction coordinates. The resulting potential of mean force features a wide, flat basin that mirrors the conformational degeneracy of the peptide—i.e., a heap of compact structures of similar energy, from which the minimum representative of the α -helical state cannot be recovered. Augmenting the dimension of the model reaction coordinate by introducing another collective variable—e.g., the root mean-square deviation with respect to some reference state, partially lifts conformational degeneracy. Only by adding a third dimension to the free-energy landscape will the different conformational states accessible to the short peptide be sufficiently well materialized and distinguished without ambiguity.⁸²

An alternate route for the exploration of the degrees of freedom orthogonal to the model reaction coordinate—which can be viewed as parallel val-

leys separated by free-energy barriers generally insurmountable over timescales commonly amenable to molecular dynamics, consists in spawning replicas or walkers,⁸⁷ which are run in parallel, and exchanging information at fixed intervals throughout the simulation. Diffusion along the reaction coordinate is markedly enhanced by virtue of a selection mechanism, which clones far-reaching walkers, whereas eliminating the less effective ones. The overall convergence properties of the one-dimensional potential-of-mean-force calculation are significantly improved, as the algorithm guarantees a more homogeneous sampling of the heap of compact conformations, preserving the global minimum that corresponds to the α -helical state of the decapeptide.

A Host of Methods to Measure the Free-Energy Changes

In sharp contrast with alchemical transformations, which are under most circumstances handled using either free-energy perturbation^{8,9} or thermodynamic integration¹¹—or close variants thereof, geometrical transformations have been the object of the development of a large variety of collective-variable-based algorithms of relative merits and drawbacks. These algorithms are disseminated in the four categories, or classes of methods, outlined earlier in this review.

A popular approach devised 35 years ago and coined umbrella sampling⁸⁸ epitomizes the fundamental concept of importance sampling, a variance-reduction technique central to many algorithms aimed at favoring certain regions of configurational space that are important for the estimation of the free-energy change despite a possible low probability of being visited in a molecular dynamics simulation.⁵ Strictly speaking, in umbrella sampling free-energy calculation, an external potential is introduced, which, within a range of the reaction pathway—in most applications, umbrella sampling is combined with staging⁴¹ for improved numerical efficiency—exactly compensates the variation of the free energy along the collective variable of interest, so that diffusion proceeds on an apparently flat free-energy surface. The dilemma is that for any qualitatively new problem, the mathematical form of the umbrella is in large measure unknown, to the extent that the modeler resorts to educated guesses, which, under a number of circumstances, may prove to be a tedious task, prompting the development of automated, self-consistent variations. Bereft of an explicit umbrella potential, stratification strategies,⁸⁹ whereby the reaction pathway is broken down into a large number of small, overlapping windows—which are

subsequently pasted together using such histogram-based algorithms as the well-known weighted histogram analysis method⁷—are often referred to as umbrella sampling.

Interestingly enough, compartmentalization of research areas, already evoked in the context of the rediscovered Bennett acceptance ratio, has another collateral consequence—the development of algorithms that are avatars of previously published methodologies. Eight years after the inception of the local-elevation method by Huber et al.,⁹⁰ shortly followed by its cousin algorithm coined conformational flooding by Grubmüller,⁹¹ Laio and Parrinello⁹² proposed a rendition of the two latter numerical schemes, which they called metadynamics. United by common roots, these approaches rely upon the introduction in molecular-dynamics simulations of a memory-dependent potential, disfavoring regions of conformational space that have already been visited.

To a large extent, adaptive biasing potential algorithms⁵ can be viewed as sequels of the local elevation⁹⁰ and conformational flooding⁹¹ strategies, wherein an increase of the bias penalizes sampling of those regions of the collective-variable space that have been amply explored. In this sense, the latter numerical schemes are distinct from the adaptive biasing force algorithm put forth by Darve and Pohorille,⁹³ which pertains to the class of gradient-based methods and approximates the mean force in regions heretofore visited,⁹⁴ thereby providing a Hamiltonian bereft of an average force acting along the model reaction coordinate of interest. Contrary to the original umbrella sampling method, the adaptive biasing force algorithm, notably in its unconstrained molecular dynamics version,⁸⁵ does not require any *a priori* knowledge of the topology of the free-energy landscape. One practical advantage of the adaptive biasing force algorithm and related thermodynamic-integration schemes lies in the possibility to decompose formally⁹⁵ into physically meaningful contributions the average force acting on the collective variable of interest, thus, helping dissect the nature of the interactions at play.

For the sake of completeness, the picturesquely named blue moon ensembles⁹⁴ ought to be mentioned, as yet another gradient-based alternative, albeit using constrained molecular-dynamics simulations. Different variants of the accelerated molecular dynamics algorithm^{96,97} have recently emerged as popular options for sampling rare events. Although they are not free-energy methods, per se, they can in principle be utilized to determine free-energy changes, assuming appropriate reweighting.

Potentials of Mean Force and Transport Phenomena

Potential-of-mean-force calculations⁸⁹ can be viewed as *computational tweezers* employed to dissect processes of the cell machinery. Although the format of the present review precludes an exhaustive account of the many biologically relevant processes investigated by means of such numerical simulations, a number of landmark studies, which have shaped the field, ought to be mentioned, chief among which is that of ion-channel selectivity. To understand how membrane channels discriminate between ions, elucidating how attractive ion-channel and repulsive ion-ion interactions balance out is crucial. Resorting to stratification simulations, Bernèche and Roux⁹⁸ tackled 12 years ago the mechanism of potassium-ion conduction in KcsA, for which the structure of the pore had been resolved a couple of years earlier. Their multi-ion potential-of-mean-force calculations brought to light two prevailing states, wherein either two or three potassium ions occupy the selectivity filter. The highest free-energy barrier toward ion permeation was measured to be on the order of 2–3 kcal/mol, thus, suggesting that conduction in the channel is limited by self-diffusion.

Three years later, Allen et al.,⁹⁹ revisited ion conduction through the paradigmatic gramicidin A in a one-hundred-nanosecond biased molecular-dynamics simulation. To gain additional insight into the mechanism of potassium-ion permeation, they generated a two-dimensional free-energy surface along both longitudinal and radial directions. From the resulting free-energy landscapes, corrected to account for the absence of explicit induction phenomena in the simulations, single-channel conductance was measured and shown to be of unprecedented agreement with experiment. Decomposition of the potential of mean force highlights a noteworthy propensity of the water wire to stabilize potassium ions and a very specific role played by the orientation of the constituent water molecules on conductivity.

In a synergistic theoretical and experimental investigation,¹⁰⁰ Beckstein and Sansom made use of the recent dynamic importance-sampling algorithm¹⁰¹ to address conformational transitions in the sodium-hydantoin transporter Mhp1. On the basis of large ensembles of trajectories representative of the alternate conformational states, they examined the rotation of two rigid-body domains coupled with the bending of two transmembrane helices, which underlie the opening and closing of the carrier, prefacing the diffusion of substrates through the latter.

More recently, Lau and Roux¹⁰² addressed the question of ligand-induced conformational changes in ionotropic glutamate receptors. Resorting to umbrella-sampling simulations, they mapped the conformational free-energy landscape of the ligand-binding domain for a set of substrates and determined the standard binding affinities for this set of ligands employing the theoretical framework of Woo and Roux,⁷² which relies on the addition of geometrical restraints. The structural information inferred from these free-energy calculations provides a solid and general basis for explaining the effect of partial agonists and understanding the function of a wide class of ligand-activated receptors and channels.

Returning to the problem of membrane insertion of nascent proteins, Gumbart, Teo, Roux, and Schulten investigated the thermodynamics and kinetics of protein segments conveyed by a bacterial translocon.¹⁰³ At the frontiers of what free-energy calculations can achieve today, they showed using umbrella-sampling simulations that diffusion of the nascent chain is guided by local free-energy differences between the channel and the lipid environment. Their time-dependent probability of membrane insertion is suggestive of a two-state mechanism, wherein the thermodynamics of partitioning is underlain by the kinetics of the process at hand.

Potentials of Mean Force and Recognition and Association Phenomena

Potential-of-mean-force calculations also play a considerable role in understanding how chemical species recognize and associate. One might argue that they actually provide a more satisfactory picture of these phenomena than alchemical free-energy calculations, because they follow the molecules of interest from their free, unbound state to the native, bound state of the dimeric complex. This point of view is obviously misleading, as the model reaction coordinate followed to describe association bears certain arbitrariness and, in all likelihood, unless proven otherwise—e.g., by means of a committor analysis, does not correspond to a physically realistic pathway.

Still, potentials of mean force have helped deciphering intricate recognition and association phenomena of the cell machinery, complementing their thermodynamic signature with a detailed atomic description of the underlying processes. Such is the case, for instance, of the reversible association of transmembrane α -helices. Employing the adaptive biasing force method, Hénin et al.,⁹⁵ estimated the free energy of glycophorin-A α -helix dimerization in a membrane mimetic. Turning to a rudimentary hydrocarbon en-

vironment is fully justified by the stringent limitations of molecular dynamics simulations to capture the slow relaxation of the in lipid fatty tails in response to the diffusion of the helical segments therein. In addition, measurements of standard binding affinities for such transmembrane α -helices, either by analytical ultracentrifugation, or fluorescence resonance energy transfer, is ordinarily performed in a variety of detergents. It follows that theoretical and experimental 1-molar standardized dimerization free energies cannot be compared in a straightforward fashion. Yet, these potential-of-mean-force calculations illuminate distinct regimes in the sequence of events that marks recognition and association phenomena in membranes. Specifically, at large separations, the upright α -helical segments are pushed together by the solvent, a process conducive to the creation of early inter-helix contacts. This pivotal step prefaces the formation of native van der Waals contacts, concomitant with the concerted tilt of the transmembrane segments, resulting in the characteristic structural layout of the dimer.

Similar free-energy calculations, consisting in determining one-dimensional potentials of mean force, have been performed by Hsin, LaPointe, Chipot, Senes and Schulten in the context of PufX¹⁰⁴—a dimer of transmembrane helices found in the light-harvesting I-reaction center core complex of the photosynthetic apparatus of certain bacteria. Following a strategy akin to that used for glycophorin A, the dimerization affinity of PufX was measured in four *Rhodobacter* species, namely *Rhodobacter blasticus*, *capsulatus*, *sphaeroides*, and *veldkampii* resorting to adaptive-biasing-force calculations, alongside with semi-quantitative ToxCAT assays. Although theory and experiment reveal that PufX α -helices in the four *Rhodobacter* species have markedly different dimerization affinities, they also indicate that in photobacteria possessing dimeric core complexes, PufX can potentially homodimerize, when in the only specie endowed with monomeric core complexes, PufX shows little to no dimerization propensity. These results tend to suggest that the oligomerization state of the core complex in different *Rhodobacter* membranes is dictated, at least in part, by the propensity of the constituent PufX α -helical segments to homodimerize.

Measure of the standardized binding free energies of both glycophorin-A and PufX α -helices outlined above relies upon the determination of one-dimensional separation potentials of mean force—which quite optimistically assumes proper averaging of other, orthogonal degrees of freedom, in particular the intrinsic rotation of the participating

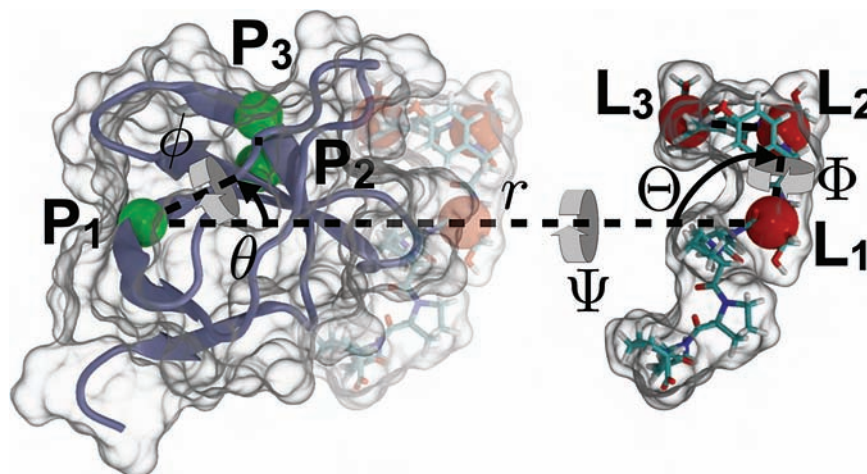


FIGURE 3 | Framework used to characterize the binding of proline-rich peptide p41 (stick representation colored by atom type) to the src homology domain 3 of the Abl kinase (secondary-structure representation).¹⁰⁵ For both the protein and the ligand, three groups of atoms are chosen arbitrarily, forming two triplets, { P1, P2, P3 } and { L1, L2, L3 }, respectively. The position of ligand p41 with respect to the protein is defined by means of the set of the spherical coordinates $\{r, \theta, \text{and } \phi\}$. Its relative orientation is expressed using the three Euler angles $\{\Theta, \Phi, \text{and } \Psi\}$.

transmembrane segments, alongside with their relative orientation. To which extent is this hypothesis legitimate remains unclear, as the membrane environment presumably reduces configurational entropy of the α -helices by restraining naturally the orientation of the latter. Such evidently would not be true in the case of two biological objects, e.g., a protein and a ligand, or two proteins, in bulk water, where they can reorient more freely, albeit over timescales not amenable to brute-force molecular dynamics. For this reason, as discussed previously in the context of alchemical transformations, it is strongly recommended to incorporate to the free-energy calculations a set of external harmonic potentials aimed at restricting the positional, orientational and conformational entropy of the complex to its native arrangement, and evaluate the cost incurred to enforce these geometrical restraints.

Gumbart et al.¹⁰⁵ have shown that a complete determination of protein-ligand standard binding free energies can be performed based on the computation of a series of potentials of mean force. The association constant is restated as a product of ratios of configurational integrals, which can be evaluated sequentially from the one-dimensional profiles delineating the free-energy change along the relevant conformational, positional, and orientational degrees of freedom, in the bound and the unbound states of the complex. In a proof of concept, they have examined the binding of a proline-rich peptide binding to the Src homology domain 3 of tyrosine kinase Abl, comparing the relative advantages and disadvantages of fully alchemical and geometrical free-energy calculations to

address the long-standing protein-ligand problem (see Figure 3). The experimental binding free energy, equal to -7.94 kcal/mol, is well reproduced by the two computational strategies, with -7.7 kcal/mol for the alchemical route and -7.8 kcal/mol for the alternate geometrical route. Although the two computational strategies yield equally reliable free-energy estimates, the potential-of-mean-force route appears somewhat cheaper, compared to its perturbative counterpart. On the other hand, the two routes are not strictly speaking interchangeable. Although geometrical free-energy calculations are well suited for ligands bound to the surface of the protein, they are expected to fail when the substrate is deeply ensconced in the binding pocket of the latter, on account of the many slowly relaxing degrees of freedom to which the low-collectivity variable formed by the protein-ligand distance is coupled. In this event, turning to alchemical free-energy calculations is probably preferable.

What About Nonequilibrium Work Computer Experiments?

An alternate route to explore rare events spanning timescales not amenable to molecular-dynamics simulations consists in driving the system out of equilibrium. In steered¹⁰⁶ or force-probe⁷⁴ molecular-dynamics simulations, an external force is imposed along a chosen direction of Cartesian space, causing the system to drift. In comparison with equilibrium geometrical free-energy calculations like those relying upon an adaptive biasing force, steered molecular-dynamics simulations apply either a constant or a

time-varying force, which leads to significant deviations from equilibrium conditions.

Constant-velocity steered molecular-dynamics simulations mimic the action of a mobile cantilever that acts onto a substrate in atomic-force microscopy experiments. A subset of atoms is tethered harmonically to a given point moved at constant velocity along a privileged direction of Cartesian space, which can be viewed as a coarse model reaction coordinate. In atomic-force microscopy experiments, the spring constant of the cantilever is about 10^{-3} N/m, which, on the atomic scale, translates to rather large thermal fluctuations in the position of the attached substrate. In sharp contrast, steered molecular dynamics make use of spring constants about two orders of magnitude stiffer, hence providing a sharper spatial resolution and appreciably more detailed information about the molecular interactions at play. Quite unfortunately, the timescales spanned by constant-velocity steered molecular-dynamics simulations are about a thousand to a million times shorter than those common in atomic-force microscopy experiments, which implies that high pulling speeds—typically 10^3 m/s, are required to cover large stretches of the reaction pathway. An unfortunate consequence of such a high-speed regime is the generation of a sizeable amount of irreversible work, often difficult to relate to equilibrium properties.

The groundbreaking identity put forth by Jarzynski in 1997 reconciles, however, ensembles of nonequilibrium realizations with the equilibrium free-energy change characterizing the process at hand.¹⁰ In practice, convergence of the average over realizations imposes that the simulations be carried out at near-equilibrium pulling speeds. Hummer and Szabo¹⁰⁷ employed this relationship to reconstruct potentials of mean force in the quasi-equilibrium regime of atomic-force microscopy experiments—which obviates the need of a friction coefficient, as would be the case, for instance, in a Gaussian drift scheme. A theoretical framework for combining steered molecular dynamics with the Jarzynski identity was developed in 2003 and applied to the previously mentioned toy model of decaalanine in the gas phase.⁸⁴ A maximum-likelihood extension of the Jarzynski identity, combining the statistical data accrued in forward and backward pulling experiments, similar in spirit with the Bennett acceptance ratio algorithm, was proposed in 1998 by Crooks¹⁰⁸ and subsequently, in a 2006 variant, by Maragakis et al.¹⁰⁹

The first application of the Jarzynski identity to large biological objects, specifically a membrane protein, was published in 2002.¹¹⁰ Nonequilibrium work computer experiments were performed to exam-

ine the energetics of glycerol conduction through the *Escherichia coli* aquaglyceroporin GlpF, a homotetrameric membrane channel that facilitates transport of water and small alcohols. From the time series derived from steered molecular-dynamics simulations, the potentials of mean force delineating glycerol conduction in the four channels were reconstructed. This pioneering study brings to light a number of interesting features, chief among which are the two appreciable free-energy barriers emerging at the level of the selectivity filter and prefaced by a rather deep minimum in the periplasmic vestibule, hypothesized to promote uptake of glycerol.

Six years later, Hénin et al.¹¹¹ reexamined the question of glycerol conduction through the transport facilitator GlpF, resorting to equilibrium free-energy calculations that use a time-dependent bias on a Euclidian distance. At variance with the results inferred from steered molecular-dynamics¹¹⁰ and umbrella-sampling¹¹² simulations, the revisited potential of mean force possesses very simple traits, consisting primarily of a single barrier arising in the region of the selectivity filter and the height of which agrees quantitatively with the experimental activation free energy—withstanding the choice of a simple Euclidian distance between glycerol and the center of mass of the channel projected onto the longitudinal axis of the latter, which probably constitutes a very naïve and, thus, questionable model of the true reaction coordinate. More importantly, these simulations illuminate the critical role played by intramolecular relaxation on the diffusion properties of the substrate—glycerol tumbles and isomerizes on a timescale comparable to that spanned by transport through the membrane channel. Reorientation and conformational equilibrium of the permeant are in fact a prerequisite for its recognition thorough the formation of the relevant network of hydrogen bonds, and may be viewed as a computational bottleneck in the simulation of the events.

However theoretically appealing, free-energy calculations relying upon nonequilibrium work computer experiments raise a number of problematic issues. First, the high pulling speed of constant-velocity steered molecular-dynamics simulations is prone to deform the reconstructed free-energy landscape by artificially introducing ruggedness in the latter—this effect is expected to be even more pronounced as pulling of the substrate is significantly faster than the relaxation of its surroundings. Second, a collateral consequence of the high pulling-speed regime is the generation of much irreversible work, which requires augmenting appreciably the number of realizations to ensure convergence of the Jarzynski identity¹⁰, or

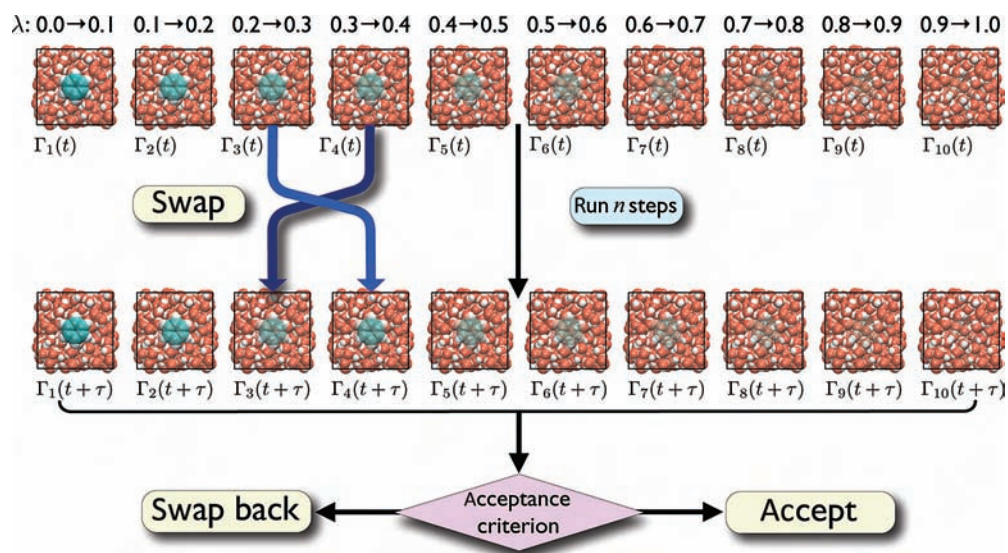


FIGURE 4 | Synoptic description of a Hamiltonian-hopping stratified free-energy perturbation calculation, wherein the hydration free energy of benzene is measured by decoupling reversibly the latter from the aqueous environment. The replicas corresponding to distinct Hamiltonians and configurations, distributed onto an array of computing cores. After a predefined number of steps, two replicas are randomly swapped. A Metropolis-Hastings-like criterion is utilized to determine whether the swap is accepted or rejected.

its Crooks equivalent.¹⁰⁸ These issues can be somewhat assuaged, notably through numerical posttreatments based upon cumulant expansions,^{84,113} or simply by reducing pulling speeds to a near-equilibrium regime—yet at the price of an appreciable computational effort. It is, therefore, not illegitimate to wonder whether there is any advantage to determine potentials of mean force from nonequilibrium work computer experiments, when equilibrium geometrical free-energy calculations employing, for instance, the adaptive-biasing force or the umbrella sampling algorithm, appear to be not only cheaper, but also more reliable.

ONGOING CHALLENGES AT THE FRONTIERS OF FREE-ENERGY CALCULATIONS

Access to continuously growing computational resources has triggered a stampede of initiatives aimed at unraveling the thermodynamics and possibly the kinetics of complex biological processes of the cell machinery—free-energy calculations are now becoming an inescapable component of the toolkit of methods available to the theoretician. As the field is propelled by exciting discoveries, which complement experiment by offering atomic-level insights rather than merely reproducing thermodynamic quantities, it also faces new challenges posed by the time and size scales being tackled. Handling the slow

degrees of freedom in the biological object of interest by means of the *computational tweezers* provided by free-energy calculations still constitutes the most prevalent roadblock thwarting measurement of reliable theoretical estimates.

Although longer brute-force simulations devour an increasingly larger number of computer cycles in an attempt to bridge the gap between time and size scales, astute, importance-sampling-based techniques are devised to explore the frustrated free-energy landscapes that underlie intricate biological phenomena. One very effective route toward this goal consists in combining in a seamless fashion families of methods to enhance, possibly accelerate sampling along privileged directions of collective-variable space. Here, families of methods refer, on the one hand, to the use of multiple copies to explore concomitantly distinct regions of phase space, e.g., parallel-tempering^{114,115} and Hamiltonian-hopping^{116,117} algorithms, and, on the other hand, the use of variance-reduction, importance-sampling strategies to improve sampling of important regions, relevant to the biological process at hand, e.g., free-energy perturbation,^{8,9} umbrella sampling,⁸⁸ and adaptive biasing force^{85,93} algorithms.

The seed of these novel approaches can be found in the juggernaut of computational power now available to the modeling community, offering a new vantage point to tackle problems hitherto unmanageable. In Hamiltonian-hopping schemes (see Figure 4),^{116,117} for instance, the different states of

a stratified free-energy perturbation calculation are handled in parallel, each simulation cell being decomposed spatially into patches distributed over the available processing units—which can result, for an appreciably sizable molecular assembly and a large number of strata, in a considerable number of cores. Swapping Hamiltonians between simulation cells at predefined times proves effective to overcome hidden barriers and enhance sampling of the slowest degrees of freedom relevant for ensemble averaging. Sampling may be further improved by feeding these slow degrees of freedom, assuming that they can be identified, with boosting potentials¹¹⁷ inferred from separate free-energy calculations—e.g., potentials of mean force. Concomitant sampling the directions of the collective variable and the time-dependent bias acting upon the latter, which is tantamount to a random walk in orthogonal space, constitutes a possible alternative to surmount hidden barriers.¹¹⁸

The theoretical framework devised to account for the change in configurational entropy upon association reweave the fabric of protein-ligand free-energy calculations,^{72,105} allowing standard binding affinities to be measured accurately. Conjugated with novel strategies for enhanced sampling, it paves the way to solving the elusive protein-protein problem. Although protein-ligand association lies at the confluence of alchemical and geometrical transformations, protein-protein recognition, and association can only be addressed following a potential-of-mean-force route. This unique choice can be understood in terms of the perturbation induced by the creation or annihilation of one of the two partner proteins and the slow response of the environment to absorb it, which readily eliminates alchemical free-energy calculations. Furthermore, the virtual impossibility to sample $8\pi^2$ of solid angle in unrestrained simulations over timescales amenable to molecular dynamics—let alone conformational rearrangements upon binding,^{119,120} imposes almost naturally to resort to appropriately chosen geometrical restraints.

Another grand challenge for free-energy calculations is to reconcile thermodynamics and kinetics through the concomitant determination of potentials of mean force and diffusivities in biased simulations. The difficulty can be construed by considering that a time-dependent bias acting on a collective vari-

able will yield, in the limit of a converged ensemble average, a Hamiltonian bereft of an average force acting along that collective variable. It follows that diffusion on the apparent flat free-energy landscape is unhindered, thereby, casting doubt on the relevance of diffusivities readily inferred from importance sampling free-energy calculations. Consistent mapping of the free-energy landscape and determination of diffusivities to yield the correct kinetic description of the process of interest, for instance by means of Bayesian inferences,^{121,122} is obviously subservient to the proper choice of a reaction-coordinate model.

CONCLUSION

Although there is a general consensus that free-energy calculations are becoming an unescapable tool of computational structural biology, thriving beyond academic walls, notably in industrial environments, the question still pervades in a skeptical community as to whether or not they have come of age to provide convincing, reliable answers. This skepticism is deeply rooted in the slow-converging nature of these numerical simulations and the formidable computational effort often required to attain acceptable results, shattering in a sense the significant discoveries that were made possible by free-energy methods. At a time when bleeding-edge brute-force, unbiased molecular dynamics struggles to bridge the still-wide gap between experimentally observed macroscopic—and numerically simulated microscopic objects on dedicated architectures endowed with thousands of computing cores, free-energy calculations represent a cogent alternative to reconcile time and size scales, assuming that the key degrees of freedom of the biological process at play are well identified. Notwithstanding robust methodological foundations, there are still many challenges for the fecund field of free-energy calculations, requiring novel developments to address profound biological questions that experiment alone has heretofore left unanswered. More than a mere matter of additional computer cycles to perform longer, bigger simulations, there is a need for original ideas to unlock the doors to new theoretical frameworks for a continuously improving methodology, sampling strategy and, to sum it all up, reliability of the free-energy estimates.

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REFERENCES

1. McCammon, JA, Gelin BR, Karplus M. Dynamics of folded proteins. *Nature* 1977, 267:585–590.
2. Alder BJ, Wainwright TE. Phase transition for a hard sphere systems. *J Chem Phys* 1957, 27:1208–1209.
3. Chipot C, Pohorille A, eds. *Free Energy Calculations. Theory and Applications in Chemistry and Biology*. Berlin, Heidelberg, New York: Springer Verlag; 2007.
4. Vergara-Jaque A, Poblete H, Lee E, Schulten K, Gonz-lez-Nilo F, Chipot C. Molecular basis of drug resistance in A/H1N1 virus. *J Chem Inf Model* 2012, 52:2650–2656.
5. Lelièvre, T, Stoltz G, Rousset M. *Free Energy Computations: A Mathematical Perspective*. London: Imperial College Press; 2010.
6. Ferrenberg AM, Swendsen RH. Optimized Monte Carlo data analysis. *Phys Rev Lett* 1989, 63:1195–1198.
7. Kumar S, Bouzida D, Swendsen RH, Kollman PA, Rosenberg JM. The weighted histogram analysis method for free energy calculations on biomolecules. I. The method. *J Comput Chem* 1992, 13:1011–1021.
8. Landau LD. *Statistical Physics*. Oxford: The Clarendon Press; 1938.
9. Zwanzig RW. High-temperature equation of state by a perturbation method. I. Nonpolar gases. *J Chem Phys* 1954, 22:1420–1426.
10. Jarzynski C. Nonequilibrium equality for free energy differences. *Phys Rev Lett* 1997, 78:2690–2693.
11. Kirkwood JG. Statistical mechanics of fluid mixtures. *J Chem Phys* 1935, 3:300–313.
12. Hessa T, Kim H, Bihlmaier K, Lundin C, Boekel J, Andersson H, Nilsson I, White SH, von Heijne G. Recognition of transmembrane helices by the endoplasmic reticulum translocon. *Nature* 2005, 433:377–381.
13. MacCallum JL, Bennett WFD, Tieleman DP. Partitioning of amino acid side chains into lipid bilayers: results from computer simulations and comparison to experiment. *J Gen Physiol* 2007, 129:371–377.
14. Johansson ACV, Lindahl E. Position-resolved free energy of solvation for amino acids in lipid membranes from molecular dynamics simulations. *Proteins* 2008, 70:1332–1344.
15. Dorairaj S, Allen TW. On the thermodynamic stability of a charged arginine side chain in a transmembrane helix. *Proc Natl Acad Sci USA* 2007, 104:4943–4948.
16. Gumbart JC, Chipot C, Schulten K. Free-energy cost for translocon-assisted insertion of membrane proteins. *Proc Natl Acad Sci USA* 2011, 108:3596–3601.
17. Shirts MR, Mobley DL, Chodera JD. Alchemical free energy calculations: ready for prime time? *Annu Rep Comput Chem*, 2007, 3:41–59.
18. Jorgensen WL, Ravimohan C. Monte Carlo simulation of differences in free energies of hydration. *J Chem Phys* 1985, 83:3050–3054.
19. Gao J, Kuczera K, Tidor B, Karplus M. Hidden thermodynamics of mutant proteins: a molecular dynamics analysis. *Science* 1989, 244:1069–1072.
20. Donnini S, Tegeler F, Groenhof G, Grubmüller, H. Constant pH molecular dynamics in explicit solvent with λ -dynamics. *J Chem Theory Comput* 2011, 7:1962–1978.
21. Goh GB, Knight JL, Brooks III, CL. Constant pH molecular dynamics simulations of nucleic acids in explicit solvent. *J Chem Theory Comput* 2012, 8:36–46.
22. van Gunsteren WF. Methods for calculation of free energies and binding constants: successes and problems. In: Van Gunsteren, WF, Weiner, PK, eds. *Computer Simulation of Biomolecular Systems: Theoretical and Experimental Applications*. The Netherlands: Escom; 1989, 27–59.
23. Jorgensen WL. Free-energy calculations: a breakthrough for modeling organic chemistry in solutions. *Acc Chem Res* 1989, 22:184–189.
24. Bash PA, Singh UC, Langridge R, Kollman PA. Free energy calculations by computer simulation. *Science* 1987, 236:564–568.
25. Shirts M, Pitera J, Swope W, Pande V. Extremely precise free energy calculations of amino acid chain analogs: comparison of common molecular mechanical force fields for proteins. *J Chem Phys* 2003, 119:5740–5761.
26. Oostenbrink C, Villa A, Mark AE, van Gunsteren WF. A biomolecular force field based on the free enthalpy of hydration and solvation: the Gromos force-field parameter sets 53A5 and 53A6. *J Comput Chem* 2004, 25:1656–1676.

27. Hermans J, Shankar S. The free energy of xenon binding to myoglobin from molecular-dynamics simulation. *Isr J Chem* 1986, 27:225–227.
28. Miyamoto S, Kollman PA. What determines the strength of noncovalent association of ligands to proteins in aqueous solution? *Proc Natl Acad Sci USA* 1993, 90:8402–8406.
29. Åqvist, J. Calculation of absolute binding free energies for charged ligands and effects of long-range electrostatic interactions. *J Comput Chem* 1996, 17:1587–1597.
30. Bash PA, Singh UC, Brown FK, Langridge R, Kollman PA. Calculation of the relative change in binding free energy of a protein–inhibitor complex. *Science* 1987, 235:574–576.
31. Cummins PL, Ramnarayan K, Singh UC, Gready JE. Molecular dynamics/free energy perturbation study on the relative affinities of the binding of reduced and oxidized NADP to dihydrofolate reductase. *J Am Chem Soc* 1991, 113:8247–8256.
32. Reddy MR, Viswanadhan VN, Weinstein JN. Relative differences in the binding free energies of human immunodeficiency virus 1 protease inhibitors: a thermodynamic cycle–perturbation approach. *Proc Natl Acad Sci USA* 1991, 88:10287–10291.
33. Simonson T, Brünger, AT. Thermodynamics of protein–peptide binding in the Ribonuclease S system studied by molecular dynamics and free energy calculations. *Biochemistry* 1992, 31:8661–8674.
34. Liu H, Mark AE, van Gunsteren WF. Estimating the relative free energy of different molecular states with respect to a single reference state. *J Phys Chem* 1996, 100:9485–9494.
35. Pohorille A, Jarzynski C, Chipot C. Good practices in free-energy calculations. *J Phys Chem B* 2010, 114:10235–10253.
36. Kofke DA, Cummings PT. Precision and accuracy of staged free-energy perturbation methods for computing the chemical potential by molecular simulation. *Fluid Phase Equil* 1998, 150:41–49.
37. Bennett CH. Efficient estimation of free energy differences from Monte Carlo data. *J Comp Phys* 1976, 22:245–268.
38. McDonald IR, Singer K. Machine calculation of thermodynamic properties of a simple fluid at supercritical temperatures. *J Chem Phys* 1967, 47:4766–4772.
39. Lu N, Kofke DA, Woolf TB. Improving the efficiency and reliability of free energy perturbation calculations using overlap sampling methods. *J Comput Chem* 2004, 25:28–39.
40. Shirts MR, Chodera JD. Statistically optimal analysis of samples from multiple equilibrium states. *J Chem Phys* 2008, 129:124105.
41. Valleau JP, Card DN. Monte Carlo estimation of the free energy by multistage sampling. *J Chem Phys* 1972, 57:5457–5462.
42. van Duijneveldt S, Frenkel D. Computer-simulation study of free-energy barriers in crystal nucleation. *J Chem Phys* 1992, 96:4655–4668.
43. Kullback S, Leibler R. On information and sufficiency. *Ann Math Stat* 1951, 22:79–86.
44. Widom B. Some topics in the theory of fluids. *J Chem Phys* 1963, 39:2808–2812.
45. Beutler TC, Mark AE, van Schaik RC, Gerber PR, van Gunsteren WF. Avoiding singularities and numerical instabilities in free energy calculations based on molecular simulations. *Chem Phys Lett* 1994, 222:529–539.
46. Zacharias M, Straatsma TP, McCammon, JA. Separation-shifted scaling, a new scaling method for Lennard–Jones interactions in thermodynamic integration. *J Chem Phys* 1994, 100:9025–9031.
47. Hirono S, Kollman PA. Calculation of the relative binding free energy of 2'GMP and 2'AMP to ribonuclease T1 using molecular dynamics/free energy perturbation approaches. *J Mol Biol* 1990, 212:197–209.
48. Mark AE, van Gunsteren WF, Berendsen HJC. Calculation of relative free energy via indirect pathways. *J Chem Phys* 1991, 91:3808–3816.
49. Oostenbrink C, van Gunsteren WF. Single-step perturbations to calculate free energy differences from unphysical reference states: limits on size, flexibility, and character. *J Comput Chem* 2003, 24:1730–1739.
50. Christ CD, van Gunsteren WF. Enveloping distribution sampling: a method to calculate free energy differences from a single simulation. *J Chem Phys* 2007, 126:184110.
51. Jiao D, Golubkov PA, Darden TA, Ren P. Calculation of protein–ligand binding free energy by using a polarizable potential. *Proc Natl Acad Sci USA* 2008, 105:6290–6295.
52. Jorgensen WL, Buckner JK, Boudon S, Tirado-Rives J. Efficient computation of absolute free energies of binding by computer simulations. Application to the methane dimer in water. *J Chem Phys* 1988, 89:3742–3746.
53. Boresch S, Tettinger F, Leitgeb M, Karplus M. Absolute binding free energies: a quantitative approach to their calculation. *J Phys Chem B* 2003, 107:9535–9551.
54. Brandsdal BO, Österberg F, Almlöf M, Feierberg I, Luzhkov VB, Åqvist, J. Free energy calculations and ligand binding. *Adv Protein Chem* 2003, 66:123–158.
55. Deng Y, Roux B. Calculation of standard binding free energies: aromatic molecules in the T4 lysozyme L99A mutant. *J Chem Theor Comp* 2006, 2:1255–1273.
56. Roderinger T, Howell PL, Pomès, R. Calculation of absolute protein–ligand binding free energy using

- distributed replica sampling. *J Chem Phys* 2008, 129:155102.
57. Boyce SE, Mobley DL, Rocklin GJ, Graves AP, Dill KA, Shoichet BK. Predicting ligand binding affinity with alchemical free energy methods in a polar model binding site. *J Mol Biol* 2009, 394:747–763.
58. Jiang W, Hodoscek M, Roux B. Computation of absolute hydration and binding free energy with free energy perturbation distributed replica-exchange molecular dynamics. *J Chem Theory Comput* 2009, 5:2583–2588.
59. General IJ, Dragomirova R, Meirovitch H. New method for calculating the absolute free energy of binding: the effect of a mobile loop on the avidin/biotin complex. *J Phys Chem B* 2011, 115:168–175.
60. Tembe BL, McCammon, JA. Ligand–receptor interactions. *Comp Chem* 1984, 8:281–283.
61. Simonson T, Archontis G, Karplus M. Free energy simulations come of age: protein–ligand recognition. *Acc Chem Res* 2002, 35:430–437.
62. Udier-Blagovic M, De Tirado PM, Pearlman SA, Jorgensen WL. Accuracy of free energies of hydration using CM1 and CM3 atomic charges. *J Comput Chem* 2004, 25:1322–1332.
63. Vanommeslaeghe K, Hatcher E, Acharya C, Kundu S, Zhong S, Shim J, Darian E, Guvench O, Lopes P, Vorobyov I, et al. CHARMM general force field: a force field for drug-like molecules compatible with the CHARMM all-atom additive biological force fields. *J Comput Chem* 2010, 31:671–690.
64. Wang J, Wolf RM, Caldwell JW, Kollman PA, Case DA. Development and testing of a general amber force field. *J Comput Chem* 2004, 25:1157–1174.
65. Knight JL, Yesselman JD, Brooks, CL III. Assessing the quality of absolute hydration free energies among CHARMM-compatible ligand parameterization schemes. *J Comput Chem* 2013, 34:893–903.
66. Chipot C, Rozanska X, Dixit SB. Can free energy calculations be fast and accurate at the same time? Binding of low-affinity, non-peptide inhibitors to the SH2 domain of the src protein. *J Comput Aided Mol Des* 2005, 19:765–770.
67. Chodera JD, Mobley DL, Shirts MR, Dixon RW, Branson K, Pande VS. Alchemical free energy methods for drug discovery: progress and challenges. *Curr Opin Struct Biol* 2011, 21:150–160.
68. Roux B, Nina M, Pomès, R, Smith JC. Thermodynamic stability of water molecules in the Bacteriorhodopsin proton channel: a molecular dynamics and free energy perturbation study. *Biophys J* 1996, 71:670–681.
69. Gilson MK, Given JA, Bush BL, McCammon, JA. The statistical-thermodynamic basis for computation of binding affinities: a critical review. *Biophys J* 1997, 72:1047–1069.
70. Hermans J, Wang L. Inclusion of loss of translational and rotational freedom in theoretical estimates of free energies of binding. Application to a complex of benzene and mutant T4 lysozyme. *J Am Chem Soc* 1997, 119:2707–2714.
71. Dixit SB, Chipot C. Can absolute free energies of association be estimated from molecular mechanical simulations? The biotin–streptavidin system revisited. *J Phys Chem A* 2001, 105:9795–9799.
72. Woo HJ, Roux B. Calculation of absolute protein–ligand binding free energy from computer simulations. *Proc Natl Acad Sci USA* 2005, 102:6825–6830.
73. General IJ. A note on the standard state’s binding free energy. *J Chem Theor Comput* 2010, 6:2520–2524.
74. Grubmüller H, Heymann B, Tavan P. Ligand binding: molecular mechanics calculation of the streptavidin–biotin rupture force. *Science* 1996, 271:997–999.
75. Doudou S, Burton NA, Henchman RH. Standard free energy of binding from a one-dimensional potential of mean force. *J Chem Theor Comput* 2009, 5:909–918.
76. Stone JE, Phillips JC, Freddolino PL, Hardy DJ, Trabuco LG, Schulten K. Accelerating molecular modeling applications with graphics processors. *J Comput Chem* 2007, 28:2618–2640.
77. Shaw DE, Deneroff MM, Dror RO, Kuskin JS, Larson RH, Salmon JK, Young C, Batson B, Bowers KJ, Chao JC, et al. Anton, a special-purpose machine for molecular dynamics simulation. *SIGARCH Comput Archit News* 2007, 35:1–12.
78. Dror RO, Arlow DH, Maragakis P, Mildorf TJ, Pan AC, Xu H, Borhani DW, Shaw DE. Activation mechanism of the β 2-adrenergic receptor. *Proc Natl Acad Sci USA* 2011, 108:18684–18689.
79. Buch I, Giorgino T, De Fabritiis G. Complete reconstruction of an enzyme-inhibitor binding process by molecular dynamics simulations. *Proc Natl Acad Sci USA* 2011, 108:10184–10189.
80. Bolhuis PG, Dellago C, Chandler D. Reaction coordinates of biomolecular isomerization. *Proc Natl Acad Sci USA* 2000, 97:5877–5882.
81. E W, Ren W, Vanden-Eijnden E. Transition pathways in complex systems: reaction coordinates, isocommitator surfaces, and transition tubes. *Chem Phys Lett* 2005, 413:242–247.
82. Hénin, J, Forin G, Chipot C, Klein ML. Exploring multidimensional free energy landscapes using time-dependent biases on collective variables. *J Chem Theor Comput* 2010, 6:35–47.
83. Chipot C, Lelièvre, T. Enhanced sampling of multidimensional free-energy landscapes using adaptive biasing forces. *SIAM J Appl Math* 2011, 71:1673–1695.
84. Park S, Khalili-Araghi F, Tajkhorshid E, Schulten K. Free energy calculation from steered molecular

- dynamics simulations using Jarzynski's equality. *J Chem Phys* 2003, 119:3559–3566.
85. Hénin, J, Chipot C. Overcoming free energy barriers using unconstrained molecular dynamics simulations. *J Chem Phys* 2004, 121:2904–2914.
 86. Chipot C, Hénin, J. Exploring the free energy landscape of a short peptide using an average force. *J Chem Phys* 2005, 123:244906.
 87. Minoukadeh K, Chipot C, Lelièvre, T. Potential of mean force calculations: a multiple-walker adaptive biasing force approach. *J Chem Theor Comput* 2010, 6:1008–1017.
 88. Torrie GM, Valleau JP. Nonphysical sampling distributions in Monte Carlo free energy estimation: umbrella sampling. *J Comput Phys* 1977, 23:187–199.
 89. Roux B. The calculation of the potential of mean force using computer simulations. *Comput Phys Comm* 1995, 91:275–282.
 90. Huber T, Torda AE, van Gunsteren WF. Local elevation: a method for improving the searching properties of molecular dynamics simulation. *J Comput Aided Mol Des* 1994, 8:695–708.
 91. Grubmüller H. Predicting slow structural transitions in macromolecular systems: conformational flooding. *Phys Rev E* 1995, 52:2893–2906.
 92. Laio A, Parrinello M. Escaping free energy minima. *Proc Natl Acad Sci USA* 2002, 99:12562–12565.
 93. Darve E, Pohorille A. Calculating free energies using average force. *J Chem Phys* 2001, 115:9169–9183.
 94. Carter E, Ciccotti G, Hynes JT, Kapral R. Constrained reaction coordinate dynamics for the simulation of rare events. *Chem Phys Lett* 1989, 156:472–477.
 95. Hénin J, Pohorille A, Chipot C. Insights into the recognition and association of transmembrane α -helices. The free energy of α -helix dimerization in glycophorin A. *J Am Chem Soc* 2005, 127:8478–8484.
 96. Voter AF. Hyperdynamics: accelerated molecular dynamics of infrequent events. *Phys Rev Lett* 1997, 78:3908–3911.
 97. Maragliano L, Vanden-Eijnden E. A temperature accelerated method for sampling free energy and determining reaction pathways in rare events simulations. *Chem Phys Lett* 2006, 426:168–175.
 98. Bernèche S, Roux B. Energetics of ion conduction through the K⁺ channel. *Nature* 2001, 414:73–77.
 99. Allen TW, Andersen OS, Roux B. Energetics of ion conduction through the gramicidin channel. *Proc Natl Acad Sci USA* 2004, 101:117–122.
 100. Shimamura T, Weyand S, Beckstein O, Rutherford NG, Hadden JM, Sharples D, Sansom MSP, Iwata S, Henderson PJF, Cameron AD. Molecular basis of alternating access membrane transport by the sodium-hydantoin transporter Mhp1. *Science* 2010, 328:470–473.
 101. Perilla JR, Beckstein O, Denning EJ, Woolf TB. Computing ensembles of transitions from stable states: dynamic importance sampling. *J Comput Chem* 2011, 32:196–209.
 102. Lau AY, Roux B. The hidden energetics of ligand binding and activation in a glutamate receptor. *Nat Struct Mol Biol* 2011, 18:283–287.
 103. Gumbart JC, Roux B, Chipot C. Efficient determination of protein-protein standard binding free energies from first principles. *J Chem Theor Comput* 2013 (in revision).
 104. Hsin J, LaPointe LM, Kazy A, Chipot C, Senes A, Schulten K. Oligomerization state of photosynthetic core complexes is correlated with the dimerization affinity of a transmembrane helix. *J Am Chem Soc* 2011, 133:14071–14081.
 105. Gumbart JC, Roux B, Chipot C. Standard binding free energies from computer simulations: what is the best strategy? *J Chem Theor Comput* 2013, 9:794–802.
 106. Izrailev S, Stepaniants S, Isralewitz B, Kosztin D, Lu H, Molnar F, Wriggers W, Schulten K. Steered molecular dynamics. In: Deuffhard, P, Hermans J, Leimkuhler B, Mark AE, Skeel R, Reich S, eds. *Computational Molecular Dynamics: Challenges, Methods, Ideas. Vol. 4, Lecture Notes in Computational Science and Engineering*. Berlin: Springer Verlag; 1998, 39–65.
 107. Hummer G, Szabo A. Free energy reconstruction from nonequilibrium single-molecule pulling experiments. *Proc Natl Acad Sci USA* 2001, 98:3658–3661.
 108. Crooks GE. Nonequilibrium measurements of free energy differences for microscopically reversible Markovian systems. *J Stat Phys* 1998, 90:1481–1487.
 109. Maragakis P, Spichty M, Karplus M. Optimal estimates of free energies from multistate nonequilibrium work data. *Phys Rev Lett* 2006, 96:100602.
 110. Jensen, MØ, Park S, Tajkhorshid E, Schulten K. Energetics of glycerol conduction through aquaglyceroporin GlpF. *Proc Natl Acad Sci USA* 2002, 99:6731–6736.
 111. Hénin, J, Tajkhorshid E, Schulten K, Chipot C. Diffusion of glycerol through Escherichia coli aquaglyceroporin GlpF. *Biophys J* 2008, 94:832–839.
 112. Hub JS, de Groot BL. Mechanism of selectivity in aquaporins and aquaglyceroporins. *Proc Natl Acad Sci USA* 2008, 105:1198–1203.
 113. Colizzi F, Perozzo R, Scapozza L, Recanatini M, Cavalli A. Single-molecule pulling simulations can discern active from inactive enzyme inhibitors. *J Am Chem Soc* 2010, 132:7361–7371.
 114. Hansmann UHE. Parallel tempering algorithm for conformational studies of biological molecules. *Chem Phys Lett* 1997, 281:140–150.

115. Sugita Y, Kitao A, Okamoto Y. Multidimensional replica-exchange method for free-energy calculations. *J Chem Phys* 2000, 113:6042–6051.
116. Woods CJ, Essex JW, King MA. The development of replica-exchange-based free-energy methods. *J Phys Chem B* 2003, 107:13703–13710.
117. Jiang W, Roux B. Free energy perturbation Hamiltonian replica-exchange molecular dynamics (FEP/H-REMD) for absolute ligand binding free energy calculations. *J Chem Theory Comput* 2010, 6:2559–2565.
118. Zheng L, Chen M, Yang W. Random walk in orthogonal space to achieve efficient free-energy simulation of complex systems. *Proc Natl Acad Sci USA* 2008, 105:20227–20232.
119. Wang L, Siu SWI, Gu W, Helms V. Downhill binding energy surface of the barnase–barstar complex. *Biopolymers* 2010, 93:977–985.
120. Periole X, Knepp AM, Sakmar TP, Marrink SJ, Huber T. Structural determinants of the supramolecular organization of G protein-coupled receptors in bilayers. *J Am Chem Soc* 2012, 134:10959–10965.
121. Hummer G. Position-dependent diffusion coefficients and free energies from Bayesian analysis of equilibrium and replica molecular dynamics simulations. *New J Phys* 2005, 7:34. doi:10.1088/1367-2630/7/1/034
122. Comer J, Chipot C, González-Nilo FD. Calculating position-dependent diffusivity in biased molecular dynamics simulations. *J Chem Theor Comput* 2013, 9:876–882.